

Adaptive Catamorphisms for Weighted Operational Theories

Adaptive weighting. Assume the weighted operational-theory construction developed previously. Let $\Gamma R = \mathcal{T}(1, R)$ be the set of closed rewrites, let U be a rig of weights, and let V be a presentation language for admissible weighting catamorphisms. An element $v \in V$ determines the current interpretation of rewrite constructors. To update that interpretation after a rewrite, introduce

$$\delta : V \times \Gamma R \longrightarrow U \times V, \quad \delta(v, r) = (w(v, r), \tau(v, r)).$$

Here $w(v, r)$ is the weight assigned to r by the current catamorphism and $\tau(v, r)$ is the presentation to be used after r . The catamorphism itself need not be stateful while traversing the syntax of r : one first evaluates r using v , then updates v between small-step reductions.

There are two distinct ways to incorporate V into the dynamics. In an *internally adaptive* theory, the presentation is part of the configuration. A rewrite

$$r : p \rightsquigarrow q$$

induces

$$(p, v) \rightsquigarrow (q, \tau(v, r))$$

with weight $w(v, r)$. In an *externally adaptive* theory, by contrast, the system evolves under a single presentation shared by all branches during a step. The current v_n determines a weighted rewrite operator A_{v_n} , and a separate classical control process replaces v_n by v_{n+1} between steps. The distinction is mild for stochastic semantics but crucial for quantum interference.

Nonnegative weights

Take $U = \mathbb{R}_{\geq 0}$. For a fixed presentation v , the weighted rewrite operator is determined by

$$A_v(p, q) = \sum_{\substack{r:s(r)=p \\ t(r)=q}} w(v, r).$$

Assuming local summability,

$$\sum_{r:s(r)=p} w(v, r) < \infty,$$

one may normalize the outgoing weights to obtain transition probabilities.

Internal adaptation gives a Markov process on the enlarged state space $\Gamma P \times V$. From (p, v) , the rewrite r occurs with probability proportional to $w(v, r)$ and produces the new state $(t(r), \tau(v, r))$. Thus the future transition law may depend on the rewrite history through the evolving presentation. This is natural when the weighting policy is intended to learn, accumulate resources, change priorities, or otherwise respond to individual reductions.

External adaptation instead uses a sequence $v_0, v_1, v_2, \dots$ chosen independently of which stochastic branch is taken during a particular step. The process distribution evolves as

$$\mu_{n+1} = K_{v_n} \mu_n,$$

where K_{v_n} is the normalized transition kernel determined by v_n . This is a time-inhomogeneous Markov process. External adaptation is appropriate when a scheduler, optimization loop, or surrounding block changes the weighting policy globally between rounds rather than as a consequence of one particular rewrite.

Complex amplitudes

Now take $U = \mathbb{C}$. For each presentation v , let $\mathcal{H}$ be the Hilbert space with basis $\{|p\rangle : p \in \Gamma P\}$ and define

$$A_v|p\rangle = \sum_{r:s(r)=p} w(v, r) |t(r)\rangle.$$

Parallel rewrites with the same source and target contribute to the same basis vector and therefore interfere. Assume local summability and boundedness. Choose $\alpha_v \geq \|A_v\|$ and put $B_v = A_v/\alpha_v$, so that B_v is a contraction. Let U_v be a unitary dilation of B_v on $\mathcal{H} \otimes \mathbb{C}^2$ satisfying

$$U_v(|\psi\rangle \otimes |0\rangle) = B_v|\psi\rangle \otimes |0\rangle + |\eta_{v,\psi}\rangle \otimes |1\rangle$$

for some $|\eta_{v,\psi}\rangle$. The ancillary qubit records whether the dilated evolution remains in the block representing the original weighted rewrite operator.

In the externally adaptive quantum theory, v_n remains classical and is shared by every coherent branch. At step n , adjoin a fresh ancillary qubit $|0\rangle_n$ and apply U_{v_n} to the process register and that ancilla. After n steps, projection onto the all-zero ancillary register gives the unnormalized branch

$$\frac{A_{v_{n-1}} \cdots A_{v_1} A_{v_0}}{\alpha_{v_{n-1}} \cdots \alpha_{v_1} \alpha_{v_0}} |\psi_0\rangle \otimes |0^n\rangle.$$

Equivalently, one may postselect the fresh ancilla after every step. The presentation is updated externally between applications of the dilations, so it carries no which-path information. Consequently all rewrites arriving at the same process basis vector still interfere. If each A_{v_n} is already unitary, one may take $\alpha_{v_n} = 1$ and dispense with the dilation ancillas. Because v_n is external, its update cannot depend on which coherent rewrite “occurred” without introducing a measurement or other branch-distinguishing interaction.

Internal adaptation is different. Let $\mathcal{H}_V$ have basis states $|v\rangle$ for semantic presentation states, and define the branch-sensitive operator $\tilde{A}$ on $\mathcal{H} \otimes \mathcal{H}_V$ by

$$\tilde{A}|p, v\rangle = \sum_{r:s(r)=p} w(v, r) |t(r), \tau(v, r)\rangle.$$

This operator too must be bounded. After rescaling to a contraction it is implemented by a unitary dilation on $\mathcal{H} \otimes \mathcal{H}_V \otimes \mathbb{C}^2$, again using a fresh dilation ancilla at each small-step reduction. The V register and the dilation ancilla play distinct roles: the ancilla implements a possibly nonunitary weighted operator, whereas V stores branch-dependent future semantics.

Suppose two rewrites

$$r_1, r_2 : p \rightsquigarrow q$$

have amplitudes a_1, a_2 and update the presentation to v_1, v_2 . On the postselected dilation branch their contribution is

$$a_1|q, v_1\rangle + a_2|q, v_2\rangle.$$

If $v_1 = v_2$, the amplitudes combine to $(a_1 + a_2)|q, v_1\rangle$. If v_1 and v_2 are orthogonal presentation states, the V register retains which-path information and the histories do not interfere when only the process register is observed. More generally, the interference term is scaled by $\langle v_2 | v_1 \rangle$. Interference can return if the presentation information is later coherently erased or uncomputed so that the branches again occupy the same total state.

Because V is a language of presentations, semantically equivalent presentations should not be made distinguishable merely because their syntax differs. One may quotient V by equivalence of the catamorphisms they denote, or choose canonical representatives. What matters for interference is whether the retained adaptive state encodes distinguishable future behavior.

Summary. For $U = \mathbb{R}_{\geq 0}$, external adaptation gives a globally time-varying transition kernel, while internal adaptation gives a Markov process on processes paired with adaptive state. For $U = \mathbb{C}$, both constructions are implemented through unitary dilations when the weighted operators are not themselves unitary. External adaptation keeps V outside the quantum state and therefore preserves ordinary interference across rewrite paths. Internal adaptation makes V a quantum register; branch-dependent updates then act as which-path information unless that information is subsequently uncomputed.